

Relocate

Full System Documentation

Moving & relocation platform — Mombasa & Nairobi, Kenya

Version 0.1.0 · July 2026

[techwizh/relocationweb](#)

Relocate — Full System Documentation

Book moving vehicles in **Mombasa** and **Nairobi**, Kenya. Own-fleet model with M-Pesa payments, in-app chat, live GPS tracking, driver portal, and admin dashboard.

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Live URLs

Service	URL	Purpose
Website (public)	Your Vercel URL (e.g. <code>https://relocationweb.vercel.app</code>)	Customers, drivers, admin UI
Backend API	<code>https://relocationweb-api.onrender.com</code>	API, database, M-Pesa, uploads
Health check	<code>https://relocationweb-api.onrender.com/api/health</code>	Verify backend is running
Source code	<code>https://github.com/techwizh/relocationweb</code>	Git repository (not the live site)

Important: GitHub shows the README and code. The live website is always the **Vercel URL**.

Architecture

Production uses a **split deploy**:



Local development runs everything on one machine (`localhost:3000`) with no `API_URL` set.

Tech stack

Layer	Technology
Framework	Next.js 16 (App Router)
Language	TypeScript
Styling	Tailwind CSS 4
Database	PostgreSQL via Prisma 6
Maps	Leaflet / react-leaflet
Payments	Safaricom Daraja M-Pesa (STK Push)
Frontend host	Vercel
Backend host	Render (free tier)
Database host	Neon (free tier)
Image storage	<code>public/uploads/</code> locally; Vercel Blob in production (optional)

User roles & flows

Customer

- Visit landing page → **Book a move**
- Choose city, sub-county, ward, vehicle, date, contact details
- Pay with M-Pesa (or skip if `SKIP_MPESA_PAYMENT=true`)
- Chat with driver, track live GPS
- Optional: register at `/register` → **My account** lists all bookings

Driver

- Register at `/driver/register` (photos, license, vehicle)
- Wait for admin approval
- Log in at `/login` (Driver tab)
- **Driver portal**: toggle availability, accept assigned jobs, update status, share GPS, chat

Admin

- Log in at `/admin/login`
 - **Dashboard** — stats overview
 - **Content** — edit landing page text and upload hero/gallery images
 - **Drivers** — approve/reject applications, set internal contact phone
 - **Bookings** — assign paid bookings to approved drivers
 - **Fleet** — live map of driver GPS
-

Pages & routes

Public

Path	Description
/	Landing page (CMS-driven)
/book	New booking form
/book/[id]/pay	M-Pesa payment
/book/[id]/chat	Booking chat
/book/[id]/track	Live driver map
/book/access-denied	Shown when booking access fails
/login	Customer or driver login
/register	Customer registration
/driver/register	Driver application
/driver/register/success	Post-registration confirmation

Customer (requires login)

Path	Description
/account	My bookings, chat/track links

Driver (requires login + approved profile)

Path	Description
/driver/dashboard	Jobs, availability, GPS sharing

Admin (requires admin session)

Path	Description
/admin	Dashboard stats
/admin/content	Landing page CMS
/admin/drivers	Review driver applications
/admin/bookings	Assign drivers to bookings
/admin/fleet	Live fleet map
/admin/login	Admin login

API reference

All API routes live on **Render** in production. Vercel proxies `/api/*` to Render.

Health

Method	Path	Auth	Description
GET	<code>/api/health</code>	None	Service health check

Landing & CMS

Method	Path	Auth	Description
GET	<code>/api/landing</code>	None	Public landing page content
GET	<code>/api/admin/content</code>	Admin	Get CMS content
PUT	<code>/api/admin/content</code>	Admin	Save CMS content
POST	<code>/api/admin/upload</code>	Admin	Upload image

Bookings

Method	Path	Auth	Description
POST	<code>/api/bookings</code>	Optional customer	Create booking
GET	<code>/api/bookings/[id]</code>	Guest/customer cookie	Booking details for pay page
GET	<code>/api/bookings/[id]/access</code>	Booking access	Access check for chat/track
POST	<code>/api/bookings/[id]/pay</code>	Booking access	Initiate M-Pesa STK Push
GET	<code>/api/bookings/[id]/payment-status</code>	Booking access	Poll payment status
GET	<code>/api/bookings/[id]/messages</code>	Booking access	List chat messages
POST	<code>/api/bookings/[id]/messages</code>	Booking access	Send chat message
GET	<code>/api/bookings/[id]/location</code>	Booking access	Latest driver GPS for booking

M-Pesa

Method	Path	Auth	Description
POST	<code>/api/mpesa/callback</code>	Safaricom	Payment webhook (Render URL only)

Customer auth

Method	Path	Auth	Description
POST	/api/customer/register	None	Register
POST	/api/customer/login	None	Login
POST	/api/customer/logout	Customer	Logout
GET	/api/customer/me	Customer	Current user
GET	/api/customer/bookings	Customer	My bookings list

Driver auth & portal

Method	Path	Auth	Description
POST	/api/drivers/register	None	Driver application
POST	/api/driver/login	None	Login
POST	/api/driver/logout	Driver	Logout
GET	/api/driver/me	Driver	Current driver user
GET	/api/driver/dashboard	Driver	Jobs + profile for portal
POST	/api/driver/availability	Driver	Toggle available/offline
POST	/api/driver/location	Driver	Submit GPS coordinates
POST	/api/driver/jobs/[bookingId]/status	Driver	Advance job status

Admin

Method	Path	Auth	Description
POST	/api/admin/login	None	Admin login
POST	/api/admin/logout	Admin	Logout
GET	/api/admin/stats	Admin	Dashboard counts
GET	/api/admin/bookings	Admin	Bookings + approved drivers
POST	/api/admin/bookings/[id]/assign	Admin	Assign driver to booking
GET	/api/admin/drivers	Admin	Driver list (?status=PENDING)
POST	/api/admin/drivers/[id]/review	Admin	Approve/reject driver
PATCH	/api/admin/drivers/[id]/contact	Admin	Set driver internal phone
GET	/api/admin/fleet	Admin	Live driver locations

Database schema

PostgreSQL managed by Prisma. Key models:

Model	Purpose
User	Customers, drivers, admins (role enum)
DriverProfile	Driver status, availability, photos
Vehicle	Driver vehicle details
Booking	Move details, status, pricing
Payment	M-Pesa payment records
ChatMessage	Per-booking chat
LocationUpdate	Driver GPS history
LandingPageContent	CMS JSON for home page

Booking status flow



Driver advances: **Assigned** → **En route** → **Loading** → **In transit** → **Delivered**

Vehicle types & base prices (KES)

Type	Base price
Van	3,500
Pickup	5,500
Canter (3-ton)	9,000
Lorry (7-ton+)	15,000

Cities

- **Mombasa** — Mvita, Nyali, Kisauni, Likoni, Changamwe, Jomvu
- **Nairobi** — Westlands, Dagoretti, Lang'ata, Kibra, etc.

Authentication & sessions

All sessions use **httpOnly cookies** signed with `ADMIN_SESSION_SECRET`.

Cookie	Role	Max age
<code>relocate_admin_session</code>	Admin	8 hours
<code>relocate_customer_session</code>	Customer	30 days
<code>relocate_driver_session</code>	Driver	7 days
<code>relocate_booking_{id}</code>	Guest booking access	90 days

Split deploy rule: `ADMIN_USERNAME`, `ADMIN_PASSWORD`, and `ADMIN_SESSION_SECRET` must be **identical on Render and Vercel** so login cookies work across the proxy.

Guest booking cookies are set when a booking is created, allowing chat/track/pay without an account.

M-Pesa payments

- Customer submits booking → status `PENDING_PAYMENT`
- `/book/[id]/pay` triggers STK Push via Daraja API
- Customer enters PIN on phone
- Safaricom calls `MPESA_CALLBACK_URL` on **Render**
- Booking status → `PAID`; customer redirected to chat

Sandbox testing

Setting	Value
<code>MPESA_ENV</code>	<code>sandbox</code>
Test phone	<code>254708374149</code>
Test PIN	<code>174379</code>
Shortcode	<code>174379</code>

Set `SKIP_MPESA_PAYMENT=true` to bypass payment during testing (booking goes straight to `PAID`).

Production (live money)

- Complete Safaricom **Go Live** on [Daraja portal](#)
- Set `MPESA_ENV=production` on **Render**
- Use live shortcode, passkey, consumer key/secret
- Set `MPESA_CALLBACK_URL=https://relocationweb-api.onrender.com/api/mpesa/callback`
- Set `SKIP_MPESA_PAYMENT=false`

File uploads

Environment	Storage
Local dev	<code>public/uploads/</code>
Render (with <code>BLOB_READ_WRITE_TOKEN</code>)	Vercel Blob (public URLs)
Render (without token)	<code>public/uploads/</code> (ephemeral on Render — use Blob for production)

Allowed: JPG, PNG, WebP, GIF — max 5 MB each.

Used for: driver profile/license photos, vehicle photos, admin hero/gallery images.

Environment variables

Render (backend) — required

Variable	Description
<code>DATABASE_URL</code>	Neon PostgreSQL connection string
<code>ADMIN_USERNAME</code>	Admin login username
<code>ADMIN_PASSWORD</code>	Admin login password
<code>ADMIN_SESSION_SECRET</code>	Cookie signing secret (32+ chars)
<code>SKIP_MPESA_PAYMENT</code>	<code>true</code> to skip M-Pesa in testing
<code>MPESA_ENV</code>	<code>sandbox</code> or <code>production</code>
<code>MPESA_CONSUMER_KEY</code>	Daraja consumer key
<code>MPESA_CONSUMER_SECRET</code>	Daraja consumer secret
<code>MPESA_SHORTCODE</code>	Paybill/till (174379 sandbox)
<code>MPESA_PASSKEY</code>	Daraja passkey
<code>MPESA_CALLBACK_URL</code>	Render callback URL
<code>MPESA_ACCOUNT_REFERENCE</code>	Relocate
<code>MPESA_TRANSACTION_DESC</code>	Move booking

Vercel (frontend) — required

Variable	Description
<code>API_URL</code>	Render backend URL (no trailing slash)
<code>ADMIN_USERNAME</code>	Must match Render
<code>ADMIN_PASSWORD</code>	Must match Render
<code>ADMIN_SESSION_SECRET</code>	Must match Render
<code>SKIP_MPESA_PAYMENT</code>	<code>true</code> during testing

Do not set `DATABASE_URL` **on Vercel.**

Optional

Variable	Where	Description
<code>BLOB_READ_WRITE_TOKEN</code>	Render	Vercel Blob for persistent uploads
<code>NEXT_PUBLIC_MAPBOX_TOKEN</code>	Either	Map provider (future use)

Deployment

See also: [SPLIT-DEPLOY.md](#)

Render (backend)

```
build: npm install && npx prisma generate && npx prisma migrate deploy && npm run build
start: npm start
health: /api/health
```

Vercel (frontend)

```
build: prisma generate && next build
env: API_URL + admin secrets (no DATABASE_URL)
```

Vercel `next.config.ts` rewrites `/api/:path*` → `${API_URL}/api/:path*`.

Custom domain

- Add domain in **Vercel** → Settings → Domains (website)
- Keep M-Pesa callback on **Render** URL (or proxy if using same domain)

Local development

```
cd C:\Users\ADMIN\Projects\relocate
npm install
```

Create `.env` from `.env.example` . For local DB use Neon URL or PostgreSQL:

```
DATABASE_URL="postgresql://..."
# Do not set API_URL locally
```

```
npm run db:deploy
npm run dev
```

Open <http://localhost:3000>

Useful scripts

Command	Description
<code>npm run dev</code>	Start dev server
<code>npm run build</code>	Production build
<code>npm run db:migrate</code>	Create migration (dev)
<code>npm run db:deploy</code>	Apply migrations (prod/local)
<code>npm run db:studio</code>	Prisma Studio GUI

Project structure

```
|
├── relocate/
├── ── prisma/
│   ├── schema.prisma      # Database models
│   └── migrations/        # PostgreSQL migrations
├── ── public/uploads/     # Local image uploads
├── ── src/
│   ├── app/
│   │   ├── api/          # Backend API routes
│   │   ├── admin/        # Admin dashboard pages
│   │   ├── account/      # Customer account
│   │   ├── book/         # Booking flow
│   │   ├── driver/       # Driver portal
│   │   ├── login/ register/ # Auth pages
│   │   └── page.tsx       # Landing page
│   ├── components/       # React UI components
│   └── lib/
│       ├── admin-auth.ts  # Admin session
│       ├── customer-auth.ts # Customer session
│       ├── driver-auth.ts  # Driver session
│       ├── booking-access.ts # Guest booking cookies
│       ├── mpesa.ts        # Daraja API
│       ├── prisma.ts       # Database client
│       ├── uploads.ts      # Image upload handler
│       ├── api-server.ts   # Server-side API fetch (split deploy)
│       └── api-url.ts      # API URL helper
├── ── render.yaml         # Render deploy blueprint
├── ── vercel.json         # Vercel build config
├── ── SPLIT-DEPLOY.md     # Deploy walkthrough
└── ── SYSTEM.md           # This document
```

Troubleshooting

Problem	Likely cause	Fix
GitHub shows README only	GitHub is code storage	Use Vercel URL for the website
<code>/api/health</code> fails on Render	Build failed or service sleeping	Check Render logs; wait 30–60s on free tier
Admin login fails on Vercel	Secrets don't match Render	Sync <code>ADMIN_PASSWORD</code> and <code>ADMIN_SESSION_SECRET</code>
Failed to parse URL from <code>/api/...</code>	Server fetch without origin	Fixed in latest code; redeploy Render
M-Pesa STK push fails	Wrong keys or callback URL	Check Daraja credentials; callback must be Render HTTPS URL
Image upload fails on Render	No persistent disk	Add <code>BLOB_READ_WRITE_TOKEN</code> on Render
First request very slow	Render/Neon free tier cold start	Normal; upgrade plan for always-on
Booking chat "forbidden"	No guest cookie or not paid	Complete payment first; cookies must match domain
Build fails on Vercel with Prisma	<code>DATABASE_URL</code> set on Vercel	Remove <code>DATABASE_URL</code> from Vercel env

Security checklist (before public launch)

- ☐ Change `ADMIN_PASSWORD` from default/example
- ☐ Use a long random `ADMIN_SESSION_SECRET`
- ☐ Set real M-Pesa credentials and `SKIP_MPESA_PAYMENT=false`
- ☐ Do not commit `.env` to GitHub
- ☐ Review approved drivers before they receive jobs
- ☐ Consider custom domain + HTTPS on Vercel

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